Mutual Fund Performance Analytics and Investor Insights using Python, SQL, and Power BI

Internship Capstone Report
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Abstract

This internship project focuses on the development of an end-to-end mutual fund analytics solution using Python, SQLite, and Power BI. The primary objective of the project was to transform raw mutual fund datasets into meaningful business insights through a structured analytics workflow. The project demonstrates how data engineering, financial analytics, and business intelligence can be integrated into a single solution to support investment analysis and decision-making.

The implementation began with collecting multiple datasets related to mutual funds, including historical NAV data, Assets Under Management (AUM), investor transactions, monthly SIP inflows, portfolio holdings, benchmark indices, and scheme performance. Python scripts were developed to automate data ingestion, cleaning, validation, and preprocessing, ensuring that the datasets were consistent and suitable for analysis. The cleaned data was then stored in an SQLite database, providing a structured and efficient environment for querying and analysis.

Exploratory Data Analysis (EDA) was performed to understand trends in mutual fund performance, investor behaviour, geographical distribution of investments, and industry growth. Financial performance metrics such as Daily Returns, CAGR, Alpha, Beta, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown were calculated to evaluate the performance and risk characteristics of different mutual fund schemes.

To extend the analysis beyond traditional financial metrics, advanced analytical techniques, including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling 90-Day Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation System, and Sector Concentration Analysis using the Herfindahl-Hirschman Index (HHI), were implemented. Finally, an interactive Power BI dashboard consisting of four pages was developed to visualize key findings through charts, KPI cards, and interactive slicers.

The completed project demonstrates a practical application of data analytics in the financial domain and provides a scalable framework for analysing mutual fund performance and investor behaviour.

Introduction

The mutual fund industry has experienced significant growth over the past decade due to increasing retail participation, digital investment platforms, and greater awareness of long-term wealth creation. Every day, mutual fund companies generate large volumes of financial and transactional data, including Net Asset Value (NAV), Assets Under Management (AUM), investor transactions, portfolio holdings, and benchmark performance. Although this information is publicly available, extracting meaningful insights from it requires effective data engineering, analytical techniques, and visualization tools.

The purpose of this project was to develop a complete analytics pipeline capable of converting raw mutual fund datasets into meaningful business insights. Rather than analysing individual datasets in isolation, the project follows an end-to-end workflow beginning with data ingestion, followed by data cleaning, validation, database creation, exploratory analysis, financial performance measurement, advanced analytics, and interactive dashboard development.

Python was selected as the primary programming language due to its extensive ecosystem of data analysis libraries such as Pandas and NumPy. SQLite was used as the relational database because of its lightweight architecture and ease of integration with analytical workflows. Power BI was used for creating interactive dashboards that enable users to explore fund performance, investor trends, and market behaviour through dynamic visualizations.

The project not only focuses on calculating traditional financial performance metrics but also incorporates advanced analytical techniques that provide deeper insights into portfolio risk, investor behaviour, and fund concentration. By combining multiple technologies into a single workflow, the project demonstrates practical skills in data engineering, financial analytics, and business intelligence that are directly applicable to real-world fintech environments.

Objectives

The primary objective of this project was to design and implement a complete mutual fund analytics platform capable of processing raw financial data and transforming it into meaningful insights for analysis and decision-making.

The specific objectives of the project are as follows:

- Develop an automated ETL pipeline to collect and process multiple mutual fund datasets.
- Clean, validate, and standardize financial datasets using Python.
- Store processed datasets in an SQLite database for efficient querying and management.
- Perform Exploratory Data Analysis (EDA) to understand trends in mutual fund performance and investor behaviour.
- Compute key financial performance metrics including Daily Returns, CAGR, Alpha, Beta, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.
- Perform advanced analytical techniques such as Historical VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation, and Sector HHI analysis.
- Develop an interactive Power BI dashboard for visualizing investment trends and fund performance.
- Build a structured, reproducible analytics workflow that can be extended for future financial analysis projects.

Technology Stack

The project was developed using a combination of programming languages, analytical libraries, database technologies, and business intelligence tools. Each technology was selected based on its suitability for a specific stage of the analytics pipeline.

Python served as the primary programming language for implementing data ingestion, cleaning, validation, financial calculations, and advanced analytics. Its readability and extensive library support made it well suited for handling financial datasets.

Pandas was used extensively for data manipulation, preprocessing, aggregation, filtering, and merging multiple datasets. Most of the data cleaning and transformation operations were performed using Pandas DataFrames.

NumPy provided numerical computing capabilities required for statistical calculations such as percentage returns, Value at Risk (VaR), Conditional Value at Risk (CVaR), and rolling financial metrics.

SQLite was used as the project's relational database for storing cleaned datasets. It offered a lightweight and portable solution that integrated seamlessly with Python and supported efficient querying of structured data.

SQL was used to define the database schema and perform analytical queries on the stored datasets.

Matplotlib was used to generate visualizations during the exploratory data analysis and performance analytics stages.

Power BI was used to build an interactive dashboard consisting of four pages, allowing users to explore key performance indicators, fund performance, investor analytics, and SIP trends through interactive charts and slicers.

Git and GitHub were used for version control, enabling systematic tracking of project development and maintaining a structured repository.

Jupyter Notebook provided an interactive environment for performing exploratory analysis, financial metric calculations, and documenting analytical findings.

Dataset Description

The project was developed using multiple datasets that together represent different aspects of the mutual fund industry. Instead of relying on a single source of information, the project integrates data related to fund details, historical NAV values, investor transactions, portfolio holdings, benchmark indices, and industry statistics. This approach allowed the analytical pipeline to generate insights from both fund-level and investor-level perspectives. The Fund Master dataset served as the primary reference table and contained information such as AMFI codes, fund house names, scheme names, fund categories, expense ratios, launch dates, and benchmark indices. This dataset was used throughout the project to identify and relate individual mutual fund schemes.

The NAV History dataset contained daily Net Asset Value (NAV) records for multiple mutual fund schemes over several years. This dataset formed the foundation for calculating daily returns and financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown. The Assets Under Management (AUM) dataset provided historical information regarding the total assets managed by different fund houses. This dataset was analysed to understand industry growth trends and compare the market presence of various asset management companies.

The Monthly SIP Inflows dataset contained monthly Systematic Investment Plan (SIP) investment amounts and active SIP account statistics. This information was useful for analysing retail investor participation and observing investment trends over time. The Category Inflows dataset recorded monthly net inflows into different mutual fund categories such as Large Cap, Mid Cap, Small Cap, and Flexi Cap. This enabled comparison of investor preferences across different investment categories.

The Industry Folio Count dataset included the number of investor folios across various mutual fund segments. This dataset was used to analyse industry growth and changes in investor participation. The Scheme Performance dataset contained pre-calculated financial indicators including Alpha, Beta, Sharpe Ratio, Sortino Ratio, annual returns, expense ratios, AUM, and Morningstar ratings. These values were used for fund comparison and recommendation. The Investor Transactions dataset contained simulated investor transactions, including SIP, Lumpsum, and Redemption transactions, together with demographic information such as state, city, age group, gender, income level, and payment method. This dataset enabled investor behaviour analysis and cohort studies.

The Portfolio Holdings dataset contained stock-level portfolio allocations for each mutual fund scheme. Sector-wise allocation percentages were later used to calculate the Herfindahl-Hirschman Index (HHI) for measuring portfolio concentration. Finally, the Benchmark Indices dataset provided historical benchmark values, which were useful for comparing fund performance with market performance during financial analysis.

ETL Pipeline

A structured Extract, Transform, and Load (ETL) pipeline was developed to automate the movement of data from raw CSV files into cleaned analytical datasets and the SQLite database. Automating this process ensured consistency, reduced manual intervention, and improved reproducibility throughout the project.

The first stage of the pipeline involved data ingestion, where all raw datasets were loaded into Python using Pandas. A dedicated ingestion script was created to automatically identify CSV files, display their structure, data types, missing values, and duplicate records. This initial inspection helped verify dataset quality before further processing.

The second stage involved data cleaning. Missing values were identified and handled appropriately, duplicate records were removed where necessary, and inconsistent data types were corrected. Date columns were converted into a standard datetime format, numerical fields were validated, and column naming conventions were standardised to maintain consistency across datasets.

The project also included an AMFI code validation step. Since AMFI codes uniquely identify mutual fund schemes, validating these identifiers ensured that records across different datasets could be accurately linked without introducing inconsistencies.

Another important component of the ETL pipeline was live NAV fetching. Python scripts were developed to retrieve the latest NAV values for selected mutual fund schemes through an external API. These datasets were automatically saved as CSV files, allowing the project to incorporate updated market information whenever required.

The final stage of the ETL pipeline involved loading the cleaned datasets into an SQLite database. The database acted as a central repository for analytical queries and simplified integration with downstream analysis tasks.

Overall, the ETL pipeline transformed multiple independent raw datasets into a structured analytical environment that supported all subsequent stages of the project.

Database Design

SQLite was selected as the database management system because it is lightweight, portable, and integrates seamlessly with Python applications. Unlike server-based database systems, SQLite stores the entire database in a single file, making it particularly suitable for academic projects and analytical workflows.

The database primarily stored three cleaned datasets: NAV History, Scheme Performance, and Investor Transactions. These datasets were loaded into separate relational tables while preserving the relationships between them using the AMFI code.

The NAV History table stored daily Net Asset Values for multiple mutual fund schemes and served as the primary source for return calculations.

The Scheme Performance table contained fund-level information including returns, Alpha, Beta, Sharpe Ratio, Sortino Ratio, Maximum Drawdown, expense ratio, AUM, and Morningstar ratings.

The Investor Transactions table stored investment activity performed by individual investors, including SIP investments, lump sum investments, redemption transactions, payment methods, demographic information, and transaction dates.

The use of SQLite allowed efficient execution of SQL queries for filtering, aggregation, and summarization while maintaining a simple project structure. It also provided an effective bridge between Python-based analysis and relational database concepts.

EDA

Exploratory Data Analysis was performed to understand the overall structure of the datasets and identify important trends before calculating financial metrics. The objective of this stage was to discover patterns, verify data quality, and generate meaningful business insights.

Historical NAV values were analysed to understand how different mutual fund schemes performed over time. Line charts were used to compare NAV movements and identify periods of consistent growth or market volatility.

Industry-wide Assets Under Management (AUM) were analysed to study the growth of the mutual fund industry over multiple years. The analysis highlighted differences between fund houses and demonstrated the increasing participation of retail investors.

Monthly SIP inflow data was examined to observe changes in investor behaviour. Time-series visualizations helped identify periods of increasing investments as well as fluctuations caused by changing market conditions.

Category-wise inflows were analysed to determine which mutual fund categories attracted the highest investments during different periods. This provided useful insights into changing investor preferences.

Investor transaction data was analysed using demographic variables including state, city, age group, gender, and income level. These analyses revealed geographical investment patterns and differences in participation across demographic groups.

Overall, the EDA phase established a strong understanding of both the financial datasets and investor behaviour, providing the foundation for the more advanced analytical techniques implemented later in the project.

Performance Analytics

Performance analytics focused on evaluating the risk and return characteristics of individual mutual fund schemes using widely accepted financial performance metrics.

Daily returns were calculated from historical NAV values using percentage changes between consecutive trading days. These daily returns formed the basis for all subsequent calculations.

Compound Annual Growth Rate (CAGR) was used to measure the annualized growth of investments over different time periods, allowing comparison between funds with different performance histories.

Alpha was calculated to estimate a fund's ability to generate returns above its benchmark after accounting for market risk. Positive Alpha values indicated outperformance relative to the benchmark.

Beta measured the sensitivity of each mutual fund relative to overall market movements. Funds with higher Beta values generally exhibited greater volatility than the market.

The Sharpe Ratio evaluated risk-adjusted performance by comparing excess returns with overall volatility. Similarly, the Sortino Ratio focused specifically on downside risk by considering only negative return deviations.

Maximum Drawdown measured the largest percentage decline from a historical peak to a subsequent trough, providing an indication of the worst-case historical loss experienced by each scheme.

Together, these metrics enabled comprehensive comparison of mutual fund performance from both return and risk perspectives.

Advanced Analytics

To extend the analysis beyond conventional financial metrics, several advanced analytical techniques were implemented.

Historical Value at Risk (VaR) at the 95% confidence level was calculated to estimate the maximum expected daily loss under normal market conditions. Conditional Value at Risk (CVaR) further measured the average loss occurring beyond the VaR threshold, providing a better understanding of downside risk.

Rolling 90-Day Sharpe Ratios were calculated to evaluate how risk-adjusted performance changed over time rather than relying on a single overall value. This analysis highlighted periods of stronger and weaker fund performance.

Investor Cohort Analysis grouped investors according to the year of their first transaction. This approach enabled comparison of investment behaviour across different investor groups and identified changes in average investment amounts over time.

SIP Continuity Analysis examined the consistency of monthly SIP investments by calculating the average gap between consecutive SIP transactions. Investors whose average investment gap exceeded thirty-five days were classified as "At-Risk," allowing identification of potentially inactive investors.

A rule-based Fund Recommendation System was developed using risk grades and Sharpe Ratios. Based on the selected risk appetite, the system recommends the top-performing mutual funds with the highest risk-adjusted returns.

Finally, Sector Concentration Analysis was performed using the Herfindahl-Hirschman Index (HHI). This metric measured portfolio diversification by analysing sector-wise investment allocations. Higher HHI values indicated greater concentration in fewer sectors, while lower values suggested more diversified portfolios.

These advanced analytical techniques added significant value to the project by providing deeper insights into investment risk, investor behaviour, and portfolio structure.

Power BI Dashboard

The final stage of the project involved developing an interactive Power BI dashboard that presented the analytical findings in a visually intuitive format.

The dashboard consisted of four pages, each focusing on a different aspect of mutual fund analytics.

The Industry Overview page displayed high-level KPIs such as total AUM, SIP inflows, number of schemes, and total investor folios. Time-series visualizations illustrated industry growth and investment trends over time.

The Fund Performance page compared mutual fund schemes using scatter plots, line charts, and performance indicators. Users could analyse returns, volatility, and risk-adjusted performance across different funds.

The Investor Analytics page focused on demographic insights, transaction behaviour, state-wise investments, age-group distributions, and investment patterns.

The SIP & Market Trends page presented monthly SIP inflows alongside broader market trends, allowing users to explore relationships between investor behaviour and market performance.

Interactive slicers enabled users to filter visualizations by fund house, scheme, category, and time period, making the dashboard suitable for exploratory analysis and business decision-making.

Key Findings

The project demonstrated that combining data engineering, financial analytics, and business intelligence provides a comprehensive understanding of mutual fund performance.

Risk-adjusted metrics such as Sharpe Ratio and Sortino Ratio revealed noticeable differences between schemes with similar returns, highlighting the importance of considering both return and risk during investment decisions.

Historical VaR and CVaR analyses showed that downside risk varied significantly across funds, particularly between equity and debt-oriented schemes.

Investor Cohort Analysis indicated changing investment behaviour across different investor groups, while SIP Continuity Analysis successfully identified investors who may be at risk of discontinuing their SIP investments.

Sector HHI calculations revealed varying levels of portfolio concentration, allowing comparison of diversification strategies across mutual funds.

Overall, the project successfully transformed raw financial datasets into meaningful insights that support investment analysis and decision-making.

Challenges

Several practical challenges were encountered during the development of the project.

Managing multiple datasets from different sources required careful validation to ensure consistent AMFI codes and compatible data formats across all files.

Handling missing values, duplicate records, and inconsistent date formats required additional preprocessing before meaningful analysis could begin.

Integrating Python scripts, SQLite databases, Jupyter notebooks, and Power BI dashboards into a single workflow also required careful planning to maintain consistency throughout the project.

Another challenge involved implementing advanced financial metrics while ensuring that the mathematical calculations remained accurate and reproducible.

Despite these challenges, each issue was resolved through systematic testing, validation, and iterative improvements to the ETL pipeline and analytical workflow.

Future Scope

The current project provides a strong foundation that can be extended in several directions.

Future versions could integrate a Streamlit web application, allowing users to access dashboards and recommendations directly through a browser.

The ETL pipeline could be scheduled to automatically fetch updated NAV data at regular intervals using task schedulers or cloud-based workflows.

Advanced portfolio optimization techniques such as Markowitz Efficient Frontier could be implemented to generate optimal investment portfolios based on risk-return trade-offs.

Monte Carlo simulations could also be incorporated to project future NAV movements under different market scenarios.

Finally, machine learning models could be explored for predicting fund performance and recommending investment strategies based on historical trends and investor preferences.

Conclusion

This project successfully demonstrated the complete lifecycle of a financial data analytics solution, beginning with raw data collection and ending with interactive business intelligence dashboards. Through the integration of Python, SQLite, SQL, Jupyter Notebook, and Power BI, an end-to-end workflow was developed that automates data processing, performs detailed financial analysis, and presents insights through effective visualizations.

The implementation of both traditional financial metrics and advanced analytical techniques provided a comprehensive evaluation of mutual fund performance while also offering insights into investor behaviour and portfolio diversification. The structured ETL pipeline, relational database design, analytical notebooks, and interactive dashboard together illustrate the practical application of data engineering and business analytics within the financial domain.

Overall, the project achieved its objectives and demonstrates the ability to design, implement, and deploy a complete analytics solution using industry-standard tools and methodologies. It also provides a scalable foundation for future enhancements involving predictive analytics, portfolio optimization, and real-time financial data processing.